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A privacy scoring framework: Automation of privacy compliance and risk evaluation with standard indicators

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ABSTRACT

Personal data have become the key to data-driven services and applications whereas privacy requirements are now strongly imposed by regulations. Meanwhile, people find it difficult to understand whether the services and applications handle personal data to comply with their agreements and regulations. Therefore, the need for privacy indicators, which summarize privacy contents as forms of privacy scoring, labels, etc., has increased to empower the users' rights by providing understandable information about privacy. For firm privacy indicators, proper criteria and methods for evaluating the level of privacy risks and compliance are required. Accordingly, this paper proposes a privacy scoring framework for services in the context of handling personal data, inspired by six standardized indicators. This paper introduces detailed information on standardized indicators and proposes privacy indicators to quantify privacy scores. Also, this paper proposes methods for evaluating privacy policy based on a set of machine learning-based hierarchical binary classifiers and processes for quantifying the level of privacy risks and compliance from privacy-related information. Through analyzing privacy policies and data access lists of more than 10,000 mobile applications on Google Play Store and investigating case studies on privacy scoring of some mobile applications, this paper shows the feasibility of the proposed framework.
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1. Introduction

With emerging big data and artificial intelligence technologies (Oussous et al., 2018; Elallid et al., 2022; Pandey et al., 2021), data now have become the key to many data-driven services. In particular, personal data that is also known as personal information or personally identifiable information draw the attention of various stakeholders since the personal data bring new value and insights that enhance the qualities of the services (Malgieri and Custers, 2018). Whereas the personal data are handled through complex value chains from data creation to consumption, the flow of data is often divided into three main phases: collection, management, and use with four different stakeholders as shown in Fig. 1 ITU-T (2020), ISO Central Secretary (2019), ISO Central Secretary (2017).

Personal data is produced from data subjects, any individuals person who can be directly or indirectly identified via personal data (*a.k.a.* service users), by using services provided by data controller and processor, which utilize personal data for providing services by utilizing personal data (*a.k.a.* service providers). In addition, personal data can be exchanged or transferred from data controller to third parties for various additional purposes with the consent of data subjects.

As results of interactions between personal data stakeholders, the three main phases are identified. When data subjects try to use a new service provided by a data controller, they first see information about service with its privacy policy for sign up. After the sign up, data subjects produces and personal data by using services and provides them to data controller (i.e., data collection phase). After that, data controllers manages the collected personal data to meet privacy compliance and the consent of data subjects (i.e., data management phase). The managed personal data are

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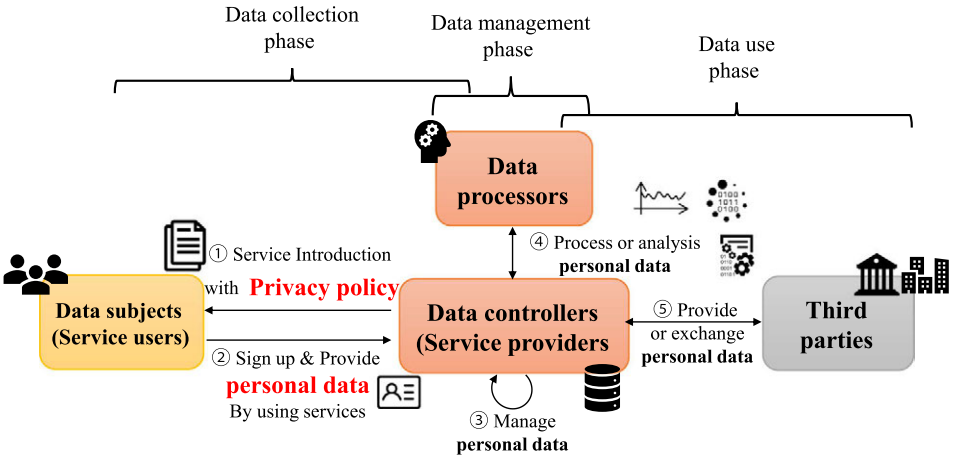


Fig. 1. An illustration of personal data stakeholders and their interactions.

processed by data controllers and/or processors to extract various information for providing services to data subjects. In addition, personal data may be provided to third parties for additional purposes (i.e., data use phase).

The data collection and use phases include the general activities of data generation, collection, analysis, and utilization, while data management phase embraces the practices that address the protection of privacy regarding the operations (including data collection and use) performed upon personal data (ISO Central Secretary, 2019). Today, the concept of personal data management is actively drawing attention in relatively recent years since personal data collection and use are now required to comply with the privacy regulations or requirements while the personal data lifecycle covers various personal data stakeholders.

As shown in Fig. 1, since privacy policy is the first item that data subjects meet for using services provided by data controllers. Recently, regulations on personal data have imposed strict requirements on the entities of personal data collection and use that need to be disclosed on privacy policies (GDPR, 2022; California Consumer Privacy Act, 2022) for empowering individual rights of data subjects in personal data ecosystem. Accordingly, personal data stakeholders should now explicitly inform the data subjects about their data collection and use in details through privacy policies. However, the data subjects seldom read the provided privacy policies since reading policies and identifying risks and choices require significant time and efforts (Schaub et al., 2015; Reidenberg et al., 2015) because the requirements of the regulations have increased the length of privacy policies (Linden et al., 2020), which non-experts often find difficulties to read (Becher and Benoliel, 2021). In other words, people usually just agree privacy policies without reading them.

To ease the issue in reading and understanding privacy policies, the concept of privacy indicator has been introduced to create synthesized or summarized privacy contents for providing more meaningful information about privacy to data subjects (Reidenberg et al., 2019; Barth et al., 2021). In particular, to provide intuitive information to service users about privacy compliance and risks of services, privacy indicators can be formed as privacy rating (e.g., grade in A, B, C, etc.), privacy score (e.g., 74 points out of 100), etc. (Reidenberg et al., 2019; Barth et al., 2021; ToS; DR, 2021; Privacy Icons, 2021; Disconnect Me, 2021; TrustArc, 2021). To produce a privacy indicator, it is important to quantify the level of privacy compliance or the risk of privacy infringement with criteria by evaluating privacy policies and other privacy factors of data controllers and processors. In particular, it is important to decide specific criteria to evaluate privacy indicators as well as data sources to quantify the criteria.

In the meantime, a privacy policy has been considered for one of the data sources for privacy indicators evaluation because it contains information about privacy and personal data handling. Therefore, various existing studies (Wilson et al., 2016; Bannihatti Kumar et al., 2020; Nejad et al., 2019; Nokhbeh Zaeem et al., 2020; Keymanesh et al., 2020; Harkous et al., 2018; Ravichander et al., 2019; Ahmad et al., 2020) have concentrated on properly analyzing texts of privacy policies to obtain information about privacy compliance and risks. To handle the texts of privacy policies, natural language processing (NLP) technologies have been applied to privacy policy analysis (Ravichander et al., 2021) in digesting verbose privacy policy documents with legal terminologies. Furthermore, for comprehensive privacy risk evaluation, various sources other than privacy policies are also investigated to identify the practices on personal data (e.g., list of requested permission, static codes analysis, etc.). In addition to the analysis of the practices on personal data, the interactions of the data controllers and processors with the data subjects are accounted into the privacy risk evaluation.

The current research and projects on privacy risk evaluation aim to analyze the data controllers' and processors' data practices and their interactions with data subjects and end users. Therefore, the previous works mostly focused on applying machine learning technologies to the field of privacy risk evaluation and enhancing performance of the analysis only with more advance and newer technologies. However, it is also important to consider not only the technologies to enhance the performance of the analysis but also provide a firm criteria to evaluate privacy indicators. In the meantime, there has not been a work that proposed privacy indicators based on international standards. To fill this gap, this study is inspired by the standards for criteria and indicators in trustworthiness evaluation of services and applications in general (ITU-T, 2017; ITU-T, 2020), which were published by International Telecommunication Union – Telecommunication (ITU-T) Standardization sector that is a *de jure* standards developing organization. This paper redefines the indicators in the context of privacy and personal data and then proposes a privacy indicator evaluation framework with the defined privacy indicators. By using the proposed framework, service users can be more empowered with intuitively understandable score about privacy compliance and risks. Moreover, service providers can improve their privacy policies and data practices for users by utilizing the results from the framework.

The contributions of this paper are as follows:

- A privacy indicator evaluation framework, which produces privacy scores, is designed that utilizes objective information on a service's data practices on personal data and subjective interactions between the service and end users to measure the privacy risks of the service in contexts of personal data collection, management, and use.
- Objective and subjective privacy indicators are defined in terms of personal data based on standards, which include diverse evaluation items to measure and quantify the privacy risks and the level of compliance caused by the personal data collection, management, and use. Moreover, the detailed criteria and evaluation methods for each privacy indicator are proposed to measure and quantify privacy indicators.
- In addition, the techniques and methods, including machine learning, to evaluate each privacy indicator are discussed in depth as the parts of the proposed evaluation framework. In particular, this paper deliberately designs the privacy policy analysis models in a hierarchical structure to identifying various data practices for the policy indicator evaluation.
- For performance evaluation, analysis results on privacy policy dataset with the proposed analysis model are introduced. Moreover, this paper analyzes privacy policies of more than 10,000 mobile applications on Google Play Store using the proposed objective privacy indicator evaluation criteria.
- Furthermore, under scenarios with different user preference on privacy, a privacy indicator of mobile applications for smart home services is evaluated as a case study with real-world data, such as privacy policy, requested permission list, ratings, and service usage information to check the effect of subjectiveness of privacy indicator.

The rest of this paper is organized as follows. In Section 2, existing studies and projects on privacy indicators and privacy risk evaluation are provided. In Section 3, the privacy indicators with the detailed criteria for measuring them and a privacy indicator evaluation framework are proposed. Performance analysis results for analyzing privacy policies and objective privacy indicators are provided. In addition, a case study for the entire proposed privacy indicator evaluation results with the proposed framework is provided as the proof of concept with two well-known mobile applications in Section 4, followed by the conclusion in Section 5.

2. Literature reviews

The goal of privacy indicators is generally considered as providing the data subjects (i.e., service users) more meaningful notice to empower them and improve the practices on personal data (Reidenberg et al., 2019; Barth et al., 2021). Accordingly, many approaches on privacy indicators that explicitly present the users' rights and their privacy risks have been widely investigated in many forms (i.e., privacy grade, score, label, certificate, etc. (ToS; DR, 2021; Privacy Icons, 2021; Disconnect Me, 2021; TrustArc, 2021)), which are often measured through the services' practices on personal data.

There also have been many works that analyze a privacy policy to identify the data practices in the policy and clearly convey them to the users of the service. Terms of Service; Didn't Read (ToS;DR) introduced a peer-reviewed process that analyzes a privacy policy to inform users of their rights and rate the services with grades (ToS;DR, 2021). Similarly, Mozilla's privacy icon initiative provided the symbolic representations about data collection or use by a service, which are affixed on the top of the service's privacy policy (Privacy Icons, 2021). Disconnect Me also introduced icons and labels to visually simplify the privacy policy that is empowered by the database from TrustArc, a privacy compliance certifying company formerly known as TRUSTe that carries out the manual analysis on the privacy policies (Disconnect Me, 2021; TrustArc, 2021). The previous works focused on various ways to represent privacy indicators in many forms. However, the previous works on privacy indicator utilized self-defined metrics, which depends on the environments of services, to measure privacy compliance and risks of services.

Meanwhile, a set of 115 online privacy policy (OPP-115) corpus was created with annotations on the practice on the personal data in Wilson et al. (2016), which has triggered many studies and projects on the automation of privacy policy analysis based on NLP technologies. The OPP-115 corpus is utilized in Bannihatti Kumar et al. (2020) to develop the models that automatically extract opt-out choices respectively from privacy policies and websites. In the meantime, privacy policies are analyzed in Nejad et al. (2019) with multi-label classifiers that are trained with the OPP-115 corpus. The classifiers categorize data-related practices in each statement, which is then analyzed through rule-based information extraction for risk measurement. In Nokhbeh Zaeem et al. (2020), a service called PrivacyCheck was developed, which provides a privacy policy summary with respect to 10 selected privacy factors for data practices in 3 risk levels by utilizing the annotated privacy policies of over 400 companies. Similarly, an approach to extract the sections that are likely to describe privacy risks as the summary of a privacy policy is proposed in Keymanesh et al. (2020) by utilizing over 150 privacy policies. In Harkous et al. (2018), deep learning-based models that classify privacy practices in privacy policies are trained with the OPP-115 and other corpora, which are further used as the solution of a service that provides summaries on privacy policies with a question–answering function, called Polisis-Pribot. Furthermore, for development of automated question–answering, crowdworkers and legal experts are utilized in Ravichander et al. (2019) and Ahmad et al. (2020) to draw the questions and answers relevant to privacy policies. Furthermore, there are works that extended the work of Wilson et al. (2016). In Zimmeck et al. (2019), a dataset was published for privacy policy analysis of mobile applications that includes 350 android application privacy policies annotated with privacy practices, and a mobile application privacy system (MAPS) was proposed that conducts privacy compliance analysis. In addition, the authors of Story et al. (2019) further examined the consistency of mobile applications' actual data accesses with data practices specified in privacy policies through NLP technologies and static code analysis as a part of MAPS. The work of OPP-115 is extended in Poplavska (2020) in the context of the European Unions' General Data Protection Regulation (GDPR). The authors identified the matches and mismatches in the annotation category of the OPP-115 corpus with GDPR and verified the alignment between their systematization, which validates OPP-115's applicability to be used in the analysis on the GDPR compliance of privacy policies. While the previous studies and projects related to OPP-115 dataset have successfully applied advanced machine learning technologies to evaluate the privacy compliance, they did not properly consider the subjectiveness of the privacy risks in the evaluation.

In the meantime, there are standardization works on indicators for evaluation on a service which insist that both objective and subjective factors need to be properly considered. In an ITU-T standard (ITU-T, 2017), direct and indirect indicators for evaluation on a service are discussed in the context of the subject's trustworthiness. The direct and indirect indicators are further extended into objective and subjective indicators for the specific domain of handling personal data in ITU-T (2020). While the objective information about a service in the contexts of privacy risks can be identified through the aforementioned research and projects, the

subjective interactions between the service and its users are required to be measured into quantitative scores. In Dagger et al. (2009), Mccaffrey et al. (2018), Haring et al. (2020), methods to transform the interactions into the generalized relationship strength between two subjects are studied.

As there has not been a work that provides an overall privacy indicator evaluation framework considering the subjectiveness of the evaluation in terms of the standardized indicators, this leaves room to improve the previous research and projects on privacy indicator evaluation. Accordingly, the scope of the paper is introducing a framework for privacy indicator evaluation in consideration of the subjectiveness of privacy risks and with the interpretation of standardized indicators in the context of privacy and personal data handling.

3. Privacy indicators evaluation framework

This section introduces the privacy indicators and a framework for evaluating the privacy risks and the level of privacy compliance to produce privacy indicators of the services handling personal data. Fig. 2 describes the proposed privacy indicator evaluation framework, which involves four main processes: 1) analysis on the privacy policy and requested data access permissions of the services, 2) evaluating the users' interactions with the services, and 3) mapping the analyzed results to the privacy indicators for evaluation, and 4) combining the evaluation of all privacy indicators at the end. The analysis on the privacy policy, data access permissions requests, and users' interactions can proceed in parallel wheres the mapping process requires all results from the previous steps to compute the privacy indicators of the services as quantitative values that are then combined into an overall privacy score. In the remainder of this section, the privacy indicators are first defined based on international standards (ITU-T, 2017; ITU-T, 2020), and the techniques and methodologies used for each process in the evaluation framework are then provided in detail.

3.1. Privacy indicator

In the evaluation on privacy risks, not only the objective information on a service's practices on personal data but also the subjective information on the interactions between the service and its users needs to be taken into account as well. Accordingly, the proposed privacy indicator evaluation framework measures a service's privacy risks as a weighted sum of the privacy indicators' scores, which are evaluated with respect to diverse objective and subjective evaluation criteria as described in Table 1.

3.1.1. Objective privacy indicator

Objective privacy indicators include the evaluation criteria which can be evaluated from objective information. Objective information implies the pieces of information that influence the same to all users. For example, the practices on personal data stated in the privacy policy of a service are objective information since the fact that the service may pursue the stated practices does not change regardless of the audiences of the privacy policy.

In that manner, the objective privacy indicators in contexts of personal data are defined as follows based on recommendations from international standards (ITU-T, 2017; ITU-T, 2020):

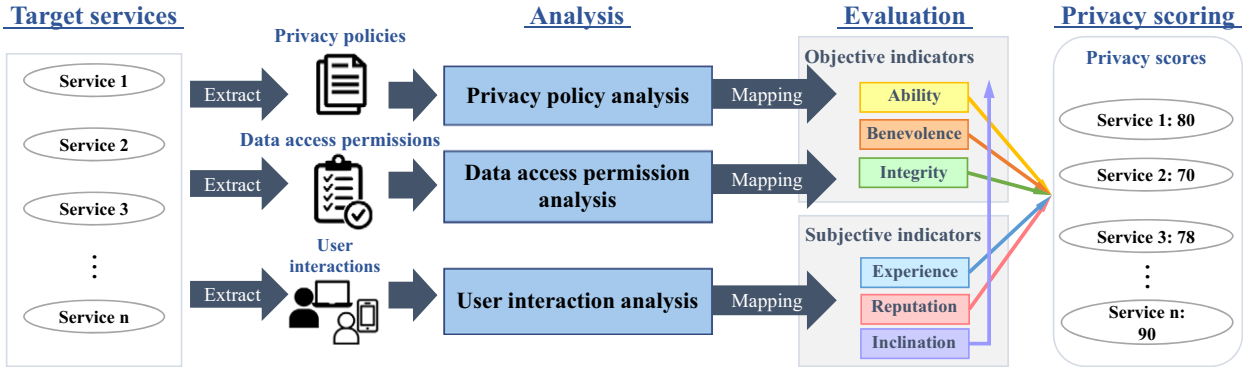


Fig. 2. Systematic overview of the proposed privacy indicator evaluation framework.

Table 1
Evaluation criteria and methods for privacy indicators.

Indicator		ID	Criteria	Method
Objective	Ability	A_1	Collecting IP or device information and online access records	Analysis on privacy policy, data access, static code, etc.
		A_2	Collecting of general personal information (i.e., contact & name)	
		A_3	Collecting of sensitive personal information (i.e., location & payment)	
		A_4	Providing security-related functions	
	Benevolence	B_1	Granting the right to choose and control personal information	
		B_2	Granting the right to access and correct personal information	
		B_3	Readability of privacy policy/ agreement terms	
		B_4	Providing contact information of the personal information manager	
	Integrity	I_1	Stating the purpose of the collection and its appropriateness	
		I_2	Sharing the collected personal information with any third party	
		I_3	Stating the type and purpose of information shared with any 3rd party	
		I_4	Notifying any change in the privacy policy or agreement terms	
Subjective	Experience	E_1	Frequency of the service use	Information from user input, external source system logs, etc.
		E_2	Duration of the service use	
	Reputation	R_1	Rating of the service	
		R_2	Number of the users who rated the service	
	Inclination	$\boldsymbol{W}$	User's preferences	

- **Ability**: the characteristics related to competency and ability in handling personal information, which may include the range and type of personal information collection, security features, etc.
- **Benevolence**: the characteristics of the service's attitude to work or act with the personal information provider, which may include whether and how to deliver the contents of the data subject's rights, such as the personal information processing or protection policy, etc.
- **Integrity**: the characteristics of the service's adherence to principles related to personal information, which may include the consistency of the personal information usage and the announced purpose, adequacy of the collection purpose, etc.

3.1.2. Subjective privacy indicator

Subjective privacy indicators include the evaluation of the accumulated interactions in both quantitative and qualitative manners. Hence, the subjective indicators can be evaluated from the intensity of the interactions between the service and its users. For example, more frequently and longer a user uses a service are intenser interactions between the service and the user in a quantitative manner. In addition, the qualitative interactions between the service and the users are stronger when more users have used the service and given high ratings.

In that manner, the subjective privacy indicators in contexts of personal data are defined as follows based on recommendations from the standards:

- **Experience**: the accumulated interactions between the user and the service, which may include relationships such as the degree of service usage in terms of frequency and duration.
- **Reputation**: the appraisals of the service's previous behavior and performance, which may include the evaluation of users.
- **Inclination**: the importance of the other privacy indicators as the weights of the privacy risks evaluation, which may differ according to the characteristics of the service and the preferences of the user.

In the remainder of this section, the approaches and methodologies for the evaluation of the defined privacy indicators are provided.

3.2. Analysis on privacy policy and requested permissions

For the evaluation of the objective privacy indicators, the sources that provide objective information on the services' data practices are utilized. The services that deal with personal data are now required by law to provide their up-to-date privacy policies in advance of personal data collection and usage. Accordingly, the privacy policies are analyzed with respect to the practices on personal data to evaluate the objective indicators. In addition, the release of the lists on the requested data access permissions is also recommended by many intermediate service distributors or can be obtained by analyzing system logs and static codes of the services. Hence, the information on the requested data access is also utilized to take consider the consistency between the privacy policy and data access in the evaluation.

3.2.1. Practices on personal data specified in privacy policy

In privacy policy analysis, various types of data practice stated in the privacy policy are identified by exploiting supervised learning with natural language processing techniques. A set of classifiers is developed to recognize the data practices from privacy policies. The classifiers are trained with the OPP-115 dataset, which is the most well-known dataset for analysis on the data practices stated in privacy policies (Wilson et al., 2016). The OPP-115 dataset provides annotations describing the data practices specified by statements in the privacy policies, which enables the transformation of data practice identification into a classification problem. The annotations were made by a group of law students in a hierarchical structure that consists of a set of data practice categories, attributes, and values. The detailed structure of the OPP-115 dataset is illustrated in Fig. 3. For our privacy indicators evaluation framework, we modified the privacy policy analysis model proposed in Sathyendra et al. (2016) since the scope of this study is introducing the overall framework for privacy indicator evaluation rather than improving the performance of the particular models in each part of the framework. In particular, we deliberately organize the privacy policy analysis models in a proposed hierarchical structure for the policy indicator evaluation as shown in Fig. 4. In Sathyendra et al. (2016), a set of classifier candidates (i.e., logistic regression, support vector machine, naïve Bayes, random forest, etc.) is investigated to determine the classifier with the best performance for extracting the users' opt-out choices stated in privacy policies. In this paper, while sharing the grounds with the policy analysis model in Sathyendra et al. (2016), a set of analysis models, each targeting to identify different data practices, are systematically designed in a hierarchical structure to recognize the data practices from privacy policies based on the proposed hierarchical classifier model.

In advance of the privacy policy analysis, the policy is formatted into the basic text and organized into sentences, and a series of

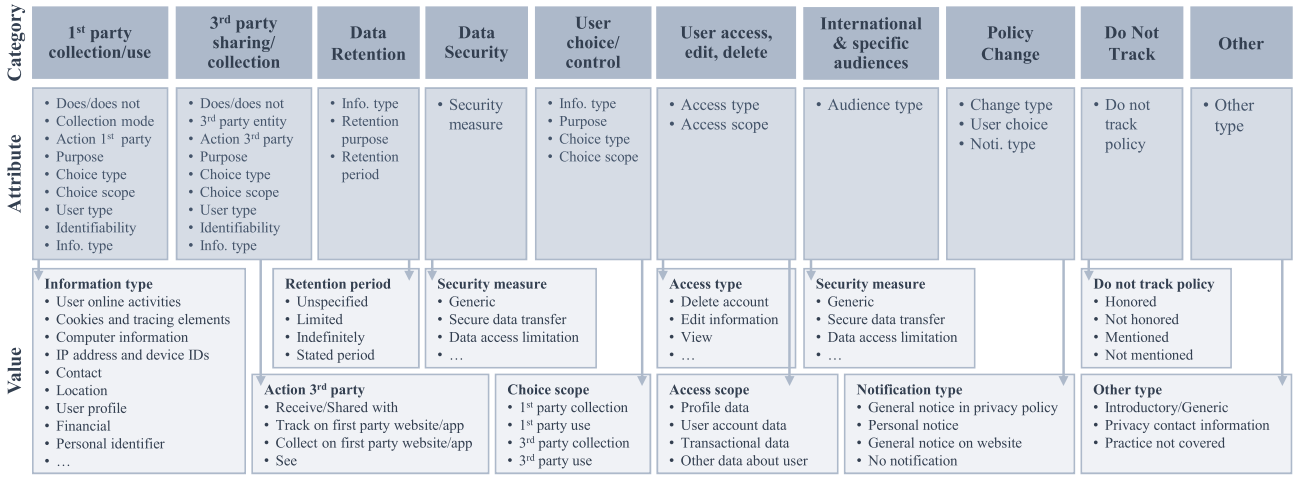


Fig. 3. Hierarchical structure of the annotations in the OPP-115 dataset.

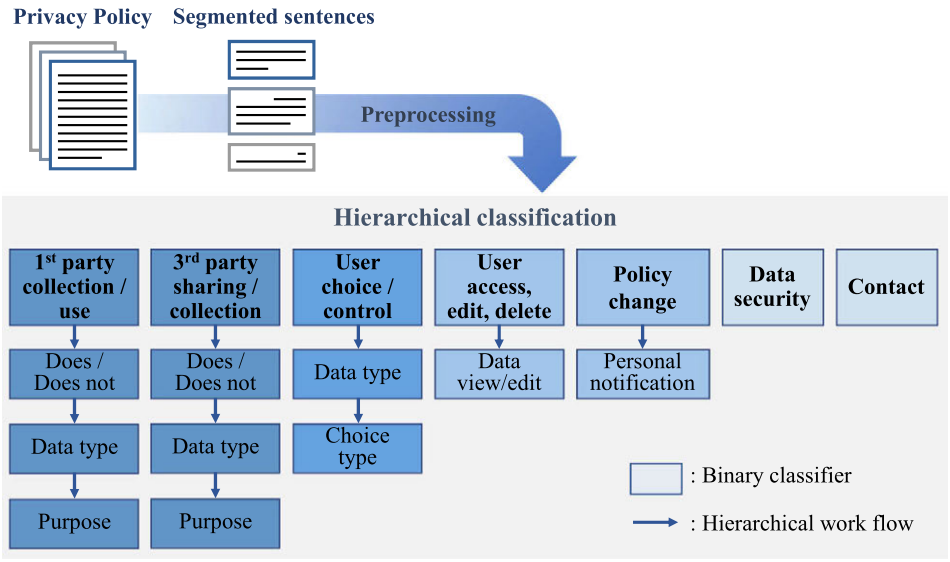


Fig. 4. Systematical structure of the proposed hierarchical classifiers for identifying data practices.

cleansing processes is applied, which removes non-letter characters, transforms all letters into lower cases, and stems words into their root forms. The sentences are then embedded with a bigram term frequency inverse document frequency (TF-IDF) vectorizer (Chowdhury, 2010), which counts the frequency of bigram bag of words in a given set of sentences and embeds the sentences into distributed vectors. To evaluate a statistical importance of a term in documents, TF-IDF vectorizer firstly measures term frequency (TF), which is the relative frequency of a term within a document, and a inverse document frequency (IDF), which is the logarithmically scaled inverse fraction of the documents that contain the term. Then, TF-IDF is calculated as the product of TF and IDF to obtain the final weight.

The classifiers take the embedded vector of each sentence to estimate the data practices that the sentence is describing. Each classifier is in charge of identifying a single type of the data practice category, attribute, or value as a binary classification, which determines whether the given sentence corresponds to each data practice category, attribute, and value. In this paper, machine learning-based classifiers are used for the classification of 47 selected categories, attributes, and values (e.g., "notify changes in policy", "about user access", etc.) that are relevant to the objective privacy indicators, which are listed in Fig. 6 in the following case study section. Meanwhile, other types of vectorization methods and classifiers can be alternatively used in the framework as aforementioned.

In addition, to identify the data practices with respect to their categories and attributes, the classifiers for the selected categories, attributes, and values are systematically designed in a hierarchical structure as described in Fig. 4. That is, for example, only the sentences recognized to have descriptions on first part collection are further processed to investigate that whether the sentences imply that the service does or does not collect personal data. In the same manner, only the sentences identified to inform that the service collects personal data are further analyzed to check whether the purpose of the data collection and use is stated, and so on. Consequently, the sentences in the privacy policy of a service are hierarchically examined through the classifiers.

3.2.2. Requested data access permission

Along with the privacy policy analysis, the information on the data access permissions requested by the services is additionally utilized in the evaluation of the objective privacy indicators. Many intermediate service distributors now advise the service providers to disclose a list of their requested data access permissions. Accordingly, we utilize the data access permission request information that is released by the service provider to identify the data collection that might not be recognized through the privacy policy analysis. In the meantime, other approaches to verify the data accesses of a service can be applied, such as static code analysis as in Zimmeck et al. (2017).

The analysis results of the privacy policy and the requested data access permissions are comprehensively used to evaluate the corresponding objective privacy indicators. The identified data practices and data accesses are used to measure the objective privacy indicators as scores between 0 and 1. The details of the mapping from the analyzed results to the privacy indicators and particular approaches for computing the scores are provided in the latter part of this section.

3.3. Evaluation of the users' interactions

While the objective privacy indicators are evaluated from the objective information sources provided by the service providers or acquired from the analysis on systematical work processes, the subjective privacy indicators require analysis on the interactions between the users and the services. Accordingly, the proposed framework utilizes the information on the users' service usages and ratings for subjective indicators evaluation to consider the interactions between the users and the services in both quantitative and qualitative manners. In addition, the proposed framework also takes the users' characteristics in the privacy indicator evaluations into account.

3.3.1. Usage of services

To quantitatively evaluate the users' experience with the services, each user's service usages are measured in terms of frequency and duration for a given period. The usage frequency and duration of a user in a logarithmic scale represent the generalized interaction intensity between the user and the service (Dagger et al., 2009). Accordingly, the frequency and duration of the service use of the user in a logarithmic scale are utilized for subjective privacy indicators evaluation. The information on the frequency and duration can be obtained by analyzing the system logs in the levels

of the service system and user equipment, or the proposed framework can ask for user inputs when such information cannot be automatically retrieved.

3.3.2. Rating on services

Meanwhile, the users' ratings on the services also need to be considered in the privacy indicator evaluation, which represents the qualitative interactions between the users and the services. In addition, the proposed evaluation framework considers the generalizability coefficient of the ratings on the service, which can be measured based on the number of users in a logarithmic scale (Mccaffrey et al., 2018; Haring et al., 2020). Consequently, the ratings and the number of users who gave the ratings are used for subjective privacy indicator (i.e., reputation) evaluation in the proposed framework. The rating information can be retrieved from the intermediate service distributors.

3.3.3. Users' characteristics

In addition, the proposed framework provides a means to include the users' characteristics in the privacy indicators evaluation by differentially weighing the importance of the privacy indicators for each user. Since each privacy indicator has a different importance to each user, the users may have different perspectives on the importance of the privacy indicators. Accordingly, the proposed evaluation framework takes the user inputs to set up the weights of the privacy indicators (i.e., inclination). The weights for each service also can be diverse depending on other criteria that are set by the users, such as service types, service providers, and service distributors. Furthermore, unsupervised learning techniques can be applied to group the services in the same boundary of the criteria in order to minimize the users' efforts in the preference settings.

3.4. Mapping the evaluation results to the privacy indicators

The analyzed results from the previous steps are used to evaluate their corresponding privacy indicators. All privacy indicators, except inclination, are evaluated into a real value in a range of $[0,1]$. In the meantime, the weights in the inclination privacy indicator are defined as a set of real positive values, where the sum of all weights is equivalent to 1.

3.4.1. Objective privacy indicators

The results of the analysis on privacy policies and requested data access permissions are used for objective privacy indicators evaluation. Accordingly, the categories, attributes, and values in the OPP-115 dataset that are related to data protection regulations are mapped to the corresponding privacy indicators as described in Table 1.

The evaluation items for the ability indicator of a service $\boldsymbol{A} = [A_1, A_2, A_3, A_4]$ are defined as

$$A_{k\in\{1,2,3\}} = \max\left\{0, \frac{A_k^{\text{thr}} - a_k}{A_k^{\text{thr}}}\right\}, \tag{1}$$

and

$$A_4 = \begin{cases} 1, \text{ if provided}, \\ 0, \text{ otherwise}, \end{cases} \tag{2}$$

where A_k^{thr} is the predefined threshold for all k in $\{1,2,3\}$, and a_k is the number of collected data types or requested accesses corresponding to the evaluation item A_k, which can be set as the average, minimum, or maximum of the number of the collected data types and requested access permissions as needed. Meanwhile, the evaluation item for the ability of the service to provide a security measure is evaluated as A_4.

In the meantime, the evaluation items for the benevolence indicator $\boldsymbol{B} = [B_1, B_2, B_3, B_4]$ are defined as

$$B_{k\in\{1,2,3\}} = \min\left\{1, \frac{b_k}{B_k^{\text{thr}}}\right\}, \tag{3}$$

and

$$B_4 = \begin{cases} 1, \text{ if provided}, \\ 0, \text{ otherwise}, \end{cases} \tag{4}$$

where b_1 and b_2 are respectively the numbers of choice types and access scopes stated in the privacy policy, and b_3 is the Flesch reading ease score that measures the readability of a text. Here, B_k^{thr} is the predefined threshold for all k in $\{1,2,3\}$. Meanwhile, the evaluation item for the benevolence indicator of the service in the manner of providing contact information of the personal information manager is measured as B_4.

For the integrity indicator $\boldsymbol{I} = [I_1, I_2, I_3, I_4]$, the evaluation items are defined as

$$I_k = \begin{cases} \min\left\{1, \frac{i_k}{I_k^{\text{thr}}}\right\}, \forall\ k \in \{1,3\}, \\ \max\left\{0, \frac{I_k^{\text{thr}} - i_k}{I_k^{\text{thr}}}\right\}, \forall k \in \{2\}, \end{cases} \tag{5}$$

and

$$I_4 = \begin{cases} 1, \text{ if notify}, \\ 0, \text{ otherwise}, \end{cases} \tag{6}$$

where i_1 is the number of collected data types stated with specific purposes while i_2 and i_3 are respectively the numbers of data types shared with any third party and the number of shared data types described with specific purposes of the sharing. Here, I_k is the predefined threshold of the k-th evaluation item in the integrity indicator for all k in $\{1,2,3\}$. Meanwhile, the evaluation item for the integrity indicator in the manner of the personal notification of the changes in the privacy policy is measured as I_4.

3.4.2. Subjective privacy indicators

The results of the analysis on the interactions between the users and the services are used for subjective privacy indicators evaluation. Accordingly, the usage frequency and duration of the services with the rating information are mapped to the corresponding privacy indicators as described in Table 1.

The evaluation items for the experience indicator $\boldsymbol{E} = [E_1, E_2]$ are measured from the service usage. The interaction intensities based on the usage frequency and duration are defined as

$$E_k = \min\left\{1, \frac{\alpha_k \log(e_k)}{\alpha_k \log(E_k^{\text{thr}})}\right\}, \tag{7}$$

where e_1 is the number of times that the user accessed the service during a given period, and e_2 is the duration in that the service was run on background or foreground in the user equipment. The predefined threshold for the k-th evaluation item for the experience indicator is denoted by B_k^{thr}. Here, α_k is defined as the parameter to indicate the rate to transform the frequency and duration into the intensity of the interactions that, however, vanishes in the evaluation process since both the measured value and the threshold are equally affected by the parameter.

Meanwhile, the evaluation items for the reputation indicator $\boldsymbol{R} = [R_1, R_2]$ are evaluated from the rating information which are defined as

$$R_1 = \min\left\{1, \frac{r_1}{R_1^{\text{thr}}}\right\}, \tag{8}$$

and

$$R_2 = \min\left\{1, \frac{\beta \log(r_2)}{\beta \log(R_2^{\text{thr}})}\right\}. \quad (9)$$

The ratio of the rating is given by R_1 where r_1 indicates the rating on average and, and R_1^{thr} is the predefined threshold. In the meantime, the generalizability coefficient of the rating is defined as R_2. where r_2 denotes the number of the users who gave the ratings, and R_2^{thr} is the predefined threshold. Here, β is defined as the parameter to indicate the rate to transform the ratings and number of the raters respectively into the ratio and generalizability coefficient, which do not affect the scores as in the experience indicator evaluation.

Unlike the other privacy indicators, the inclination indicator represents the importance of the other indicators, which is defined as a weight vector, such that $\boldsymbol{W} \in \mathbb{R}^n$, where $0 \preceq \boldsymbol{W} \preceq 1, \|\boldsymbol{W}\| = 1$, and $n = |\boldsymbol{A}| + |\boldsymbol{B}| + |\boldsymbol{I}| + |\boldsymbol{E}| + |\boldsymbol{R}|$. Here, $|\cdot|$ indicates the cardinality of a components (i.e., the number of elements). With the defined privacy indicators the privacy score is then computed as the inner product of the values of the privacy indicators with their corresponding weights as given by

$$S = \boldsymbol{V}, \boldsymbol{W}, \quad (10)$$

where $\boldsymbol{V} = [\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{I}, \boldsymbol{E}, \boldsymbol{R}]$.

In the meantime, the scope of this paper is to propose the overall framework for policy indicator evaluation as aforementioned rather than to develop advanced models for each part of the framework, which we are leaving the room for the future works of this study. Accordingly, it should be noted that our framework is not restricted to the particular techniques that we describe in this section, and other alternative methods and techniques can be extensively applied.

4. Performance analysis

This section provides performance analysis results on the proposed privacy indicator evaluation framework with various aspects. As the goal of this study is to propose a framework for privacy scoring based on standard indicators, the performance analysis of the proposed framework is provided in form of a proof of concept rather than a comparison of the quantified performance evaluation of machine learning technologies utilized in the proof of concept.

Firstly, the performance of the privacy policy analyzer is evaluated with different machine learning classifiers. Then, the analysis on objective privacy indicators for mobile applications are performed to see the trends of privacy policy descriptions in the real-world applications. After that, the effect of subjective privacy indicators on the privacy scoring is analyzed and discussed under scenarios with different user preferences on privacy.

4.1. Experiment setup

For performance analysis, this paper uses three types of data source (i.e., privacy policy, requested permission of mobile applications, and mobile application usage) for evaluate privacy indicators to measure privacy scores.

For privacy policy data, OPP-115 dataset, aforementioned in the previous sections, is used. The texts in OPP-115 dataset are annotated to indicate the meaning of the texts in terms of handling personal data. The policy analyzer described in the previous section is then developed from the annotations in the dataset. In particular, for a privacy policy analyzer that quantifies objective privacy indicators, a set of binary classifiers are developed with the OPP-115 dataset to identify the data practices from privacy policies. In data preprocessing for the analyzer, the privacy policies are first segmented into phrases using a sentence tokenizer provided in Natural Language Toolkit (NLTK). HyperText Markup Language (HTML) tags and stop words (i.e., 'a', 'the', 'and', etc.) in the segmented phrases are then removed. The words in the phrases are transformed into their morphological form with a word stemming provided in NLTK. The stemmed words are vectorized with TF-IDF vetorization, which is a numerical statistic that reflects the importance of a word to a document in a collection or corpus.

Moreover, to evaluate the objective indicators in terms of the privacy policy and the requested access of the mobile applications, the access permissions requested by the mobile applications are additionally utilized. The data of access permission requested by mobile applications are crawled from the Google Play Store in 2019 when the mobile applications were obligated to list their requested data accesses to launch their applications to the Google Play Store. The requested access data of more than 10,000 mobile applications were gathered, excluding the mobile applications without a valid Uniform Resource Locator (URL) link for a privacy policy. The requested access permission dataset obtained from Google Play Store includes 244 types of access permissions (e.g., view Wi-Fi connections, read historical network usage, capture video output, body sensors, etc.). Meanwhile, only 105 permission types related to users' behaviors (e.g., read call log, read your Web bookmarks and history, read battery statistics, etc.), identities (e.g., find accounts on the device, view configured accounts, read your contacts, etc.), and locations (e.g., approximate location (network-based), precise location (GPS and network-based), access extra location provider commands, etc.) are selected and mapped to the ability indicator for evaluation.

In addition, for mobile application usage data, Carat top 1,000 users long-term app usage (Carat) dataset is used to evaluate the subjective privacy indicators with respect to a user (Oliner et al., 2013). The Carat dataset provides the records of mobile application usages of more than 1,000 users over 2 years. In particular, attributes of the Carat dataset include battery level fraction, battery state (e.g., charging or not charging), number of running processes, memory state (e.g., number of active pages), operating system and version, device model, and a unique anonymized client identifier.

4.2. Evaluation of the privacy policy analyzer

For the privacy analyzer, the 70% of the 3,778 preprocessed segments in the OPP-115 dataset are used to train while the remaining 30% are used to test the performance of the classifiers. For simplicity, conventional machine learning-based classifiers are applied for privacy policy analysis, such that naïve Bayes (NB), support vector machines (SVM), and random forest (RF). The thresholds for the decision boundaries of the classifiers are determined as the values that minimize the false positive rates while the true positive rates are set to exceed 0.8. The performance of the binary classifiers in terms of classification accuracy are provided in Table 2 and Fig. 5.

The results present the average of the classification accuracy of 47 selected categories, attributes, and values that are relevant to the objective privacy indicators on the test dataset. The results show that the RF classifier outperforms the other types of classi-

Table 2
Averaged accuracy of the classifiers for privacy policy analysis.

Classifier	Recall	Precision	F_1 score	Accuracy
NB	62.5	91.1	69.0	62.5
SVM	80.0	93.0	83.2	80.0
RF	86.9	93.8	89.1	86.9

fiers, and the RF classifiers are accordingly applied in this paper for the further performance analysis. It should be noted that the proposed privacy indicator framework is not restricted to the certain type of classifier but other types and sets of classification methods described in Sathyendra et al. (2016) can be extensively applied.

Fig. 6 describes the detailed F_1 scores of the binary RF classifiers on the test dataset that are provided in Table 2 and Fig. 5. The performance of the RF classifiers differs by the data practice types since the number of statements about each data practice in the privacy policies is not even. The majority of the preprocessed OPP-115

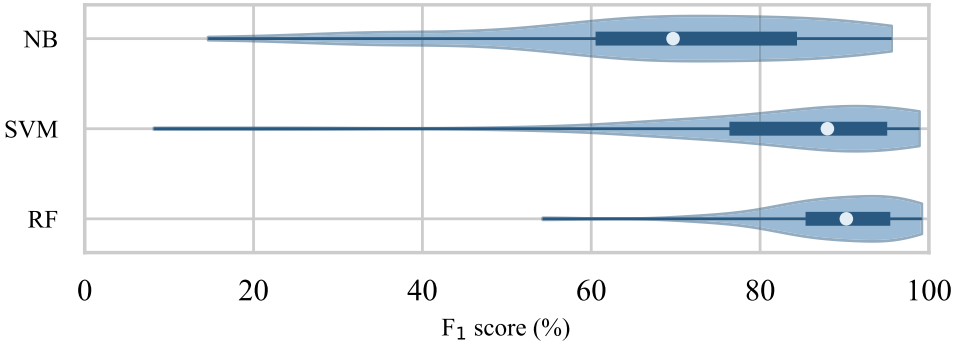


Fig. 5. F_1 scores of the classifiers for privacy policy analysis.

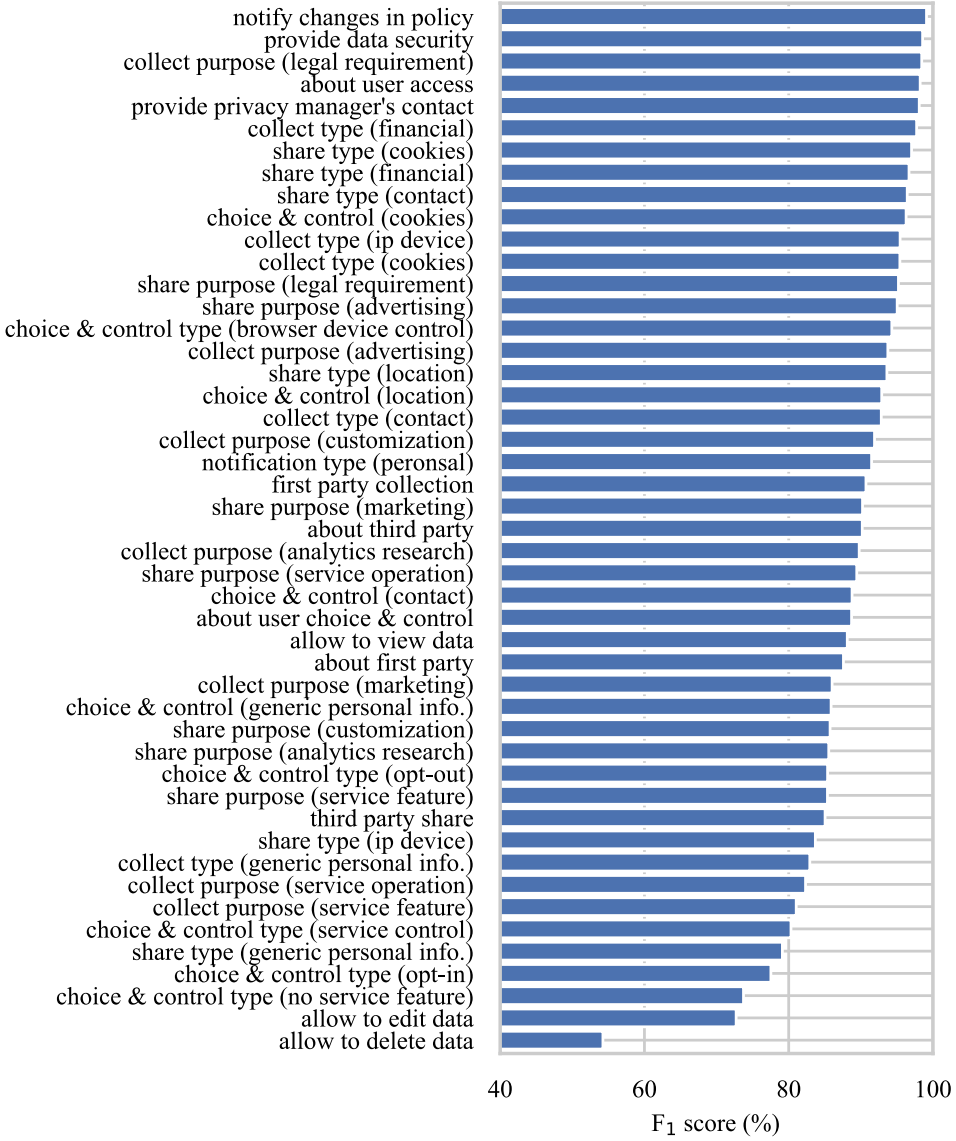


Fig. 6. F_1 scores of the binary random forest classifiers that identify each practice on personal data from privacy policies.

dataset provides information about the practices on the first party collection and third party sharing of personal data. Meanwhile, the information about user access tends to be submissive since the users' rights to access their collected personal data were not widely considered by the service providers at the time when the OPP-115 dataset was developed. Note that the OPP-115 dataset was created in the year of 2016 while the regulatory requirements about privacy policy descriptions and empowered individuals rights (e.g., choice & control, delete, etc.) has been enforced since 2018 (GDPR, 2022; California Consumer Privacy Act, 2022). As the result, the classifier that identifies the statements related to the user access, in particular data deletion, achieves comparatively poor performance.

4.3. Evaluation of objective privacy indicators

For the evaluation of the objective privacy indicators, the RF classifiers are trained with the entire OPP-115 dataset. The objective indicators of the mobile applications registered on the Google Play Store in 2019 are then evaluated with respect to the contents of their privacy policies and requested access permissions.

The trained RF classifiers are mapped to the evaluation items of each objective privacy indicator. In addition, the lists of data access permissions requested by mobile applications are also mapped to the evaluation items of the ability indicators. Fig. 7 shows the distributions of the evaluated objective indicator scores for the mobile applications registered on the Google Play Store. The mobile applications without a valid URL link for privacy policy are eliminated from the evaluation, and the objective indicator scores for 10,588 mobile applications are assessed with the standard threshold level, which is used in the following case study section. For the evaluation, the ability indicators are measured as the maximum of the number of the collected data types and requested access permission.

For the ability indicators, the score distributions of A_1, A_2, and A_3 show that the type and number of data requested by the mobile applications vary by the applications since the scores of A_1, A_2, and A_3 are evenly distributed compared to the other indicators. In the meantime, A_4 is evaluated on whether the mobile applications utilize data security technology or not, and the results show that many of the mobile applications do not explicitly mentioned the application of security technologies.

For the benevolence indicators, according to the score distributions on B_1 and B_2, some of the applications allow users to choose and control personal data while not many of them grant the users' right to access or correct the provided personal data. Meanwhile, the score distribution on B_3 implies that not many of the privacy policies have good readability for the users while the scores on B_4 implies that most of the privacy policy provide contact information of the personal information manager.

For the integrity indicators, the score distribution of I_1 implies that the mobile applications often do not clearly state the purpose of their data collection. The score distributions on I_2 and I_3 indicate that not many of the mobile applications share the collected data with third parties while the applications that share the data with third parties often do not clearly state the purpose of their data share. In addition, according to the score distribution of I_4, most of the mobile applications does not provide information on the notification for changes in the privacy policy or agreement terms.

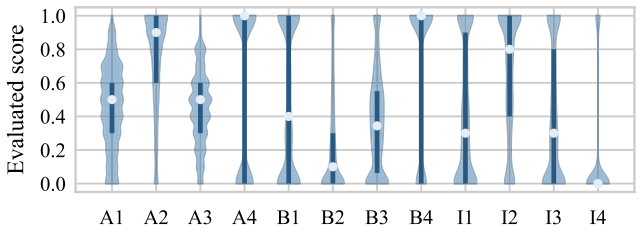


Fig. 7. Score distribution of the objective privacy indicator of mobile applications registered on the Google Play Store.

4.4. Case study: The effect of subjectiveness

This section provides a case study on the privacy indicators evaluation in a domain of mobile application considering the subjectiveness of users. The case study in this paper targets the evaluation of two well-known mobile applications for smart home, Nest and Google Home, as a proof of concept for the proposed evaluation framework. For privacy indicators evaluation of the mobile applications, the evaluation items in Table 1 are appraised with the privacy policy, requested data access permissions, rating, and usage.

The Nest mobile application uses sensors, algorithms, and the location of the users' phone to automatically control Nest thermostat, Nest secure alarm system, Nest cam, and other Nest's smart home products. The Google Home mobile application manages and controls various connected home products, such as lights, camera, thermostats, and more. These applications are chosen for the case study as it collects and uses a non-trivial amount of personal data but for limited service domains.

For the analysis, OPP-115 and requested access permission datasets are used to train machine learning models and evaluate the objective privacy indicators. In addition, the service usage information of two arbitrary users is utilized to evaluate the subjective privacy indicators of Nest and Google Home. For the detailed evaluation, the privacy indicators of Nest and Google Home are assessed with three threshold levels (i.e., mild, standard, and strict), where the thresholds in different levels reflect the sensitivity and preferences of the user to the mobile application's data practice. Here, lower the thresholds, the stricter for the privacy indicators measured as the maximum of 0 and computed values while the opposite for the other privacy indicators.

The results of the evaluation on Nest and Google Home are respectively provided in Table 3 and Table 4. While the evaluated scores both of the Nest and Google Home degrades as the thresholds are tightened, the scores of these mobile applications show different aspects.

For the subjective privacy indicators of Nest and Google Home, the evaluation on Nest and Google Home with the proposed evaluation framework is provided from the perspective of arbitrary anonymous users. The application usage information of the anonymized users is obtained from the Carat dataset, and the rating information on Google Play Store is utilized. From the Carat dataset, 67,302 accesses of an anonymous user to Nest are identified between November 2017 and February 2018, which leads to 645 accesses with approximately 10 h and 45 min of usage duration per day on average.

In the meantime, Nest has a rating of 4.2 out of 5 on average from 306,894 users on Google Play Store (Google Play Store. Nest, 2022). For another anonymous user, total 41,937 accesses to Google Home were identified between July 2017 and February 2018 from the Carat dataset, which results in 200 access with about 3 h and 20 min of usage duration per day on average. Google Home is rated 4.3 out of 5 on average by 1,357,083 users on Google Play Store (Google Play Store. Google Home, 2022).

The experience and reputation indicators are evaluated from the usage and rating information with arbitrary thresholds while the inclination indicator is set to have equal weights for all privacy indicators. Note that the value of inclination indicator also can

Table 3
Case study 1 – The privacy indicators evaluation results on Nest with different thresholds: τ_{mild}, $\tau_{standard}$, and τ_{strict}.

Category	Objective privacy indicator												Subjective privacy indicator				Total score (%)
	Ability				Benevolence				Integrity				Experience		Reputation		
	A_1	A_2	A_3	A_4	B_1	B_2	B_3	B_4	I_1	I_2	I_3	I_4	E_1	E_2	R_1	R_2	
Privacy policy	2	**5**	4	**1**	**0**	**9**	**22**	**1**	**14**	**11**	**10**	**0**	-	-	-	-	-
Access request	**10**	2	**9**	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Usage & rating	-	-	-	-	-	-	-	-	-	-	-	-	**645**	**10.75**	**4.2**	**2.84 $\times 10^5$**	-
τ_{mild}	15	15	15	1	5	5	60	1	5	15	5	1	10^2	6	4	10^5	-
Score (%)	33	67	40	100	0	100	37	100	100	27	100	0	100	100	100	100	**77**
			60				**59**				**66**			**1**		**1**	
$\tau_{standard}$	10	10	10	1	10	10	70	1	10	10	10	1	10^3	12	4.5	10^6	-
Score (%)	0	50	10	100	0	90	31	100	100	0	100	0	94	96	93	91	**68**
			40				**55**				**58**			**95**		**92**	
τ_{strict}	5	5	5	1	15	15	80	1	15	5	15	1	10^4	18	5	10^7	-
Score (%)	0	0	0	100	0	60	28	100	93	0	67	0	70	82	84	78	**57**
			25				**47**				**55**			**76**		**81**	

Table 4
Case study 2 – The privacy indicators evaluation results on Google Home with different thresholds: τ_{mild}, $\tau_{standard}$, and τ_{strict}.

Category	Objective privacy indicator												Subjective privacy indicator				Total score
	Ability				Benevolence				Integrity				Experience		Reputation		
	A_1	A_2	A_3	A_4	B_1	B_2	B_3	B_4	I_1	I_2	I_3	I_4	E_1	E_2	R_1	R_2	
Privacy policy	4	0	2	**0**	**9**	**2**	**49**	**0**	**4**	**0**	**0**	**0**	-	-	-	-	-
Access request	**6**	**2**	**7**	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Usage & rating	-	-	-	-	-	-	-	-	-	-	-	-	**200**	**3.33**	**4.3**	**1.36 $\times 10^6$**	-
τ_{mild}	15	15	15	1	5	5	60	1	5	15	5	1	10^2	6	4	10^5	-
Score (%)	60	87	53	0	100	40	82	0	80	100	0	0	100	67	100	100	**71**
			0.5				**0.55**				**0.65**			**0.83**		**1**	
$\tau_{standard}$	10	10	10	1	10	10	70	1	10	10	10	1	10^3	12	4.5	10^6	-
Score (%)	40	80	30	0	90	20	70	0	40	100	0	0	77	48	96	100	**59**
			0.38				**0.45**				**0.53**			**0.63**		**0.98**	
τ_{strict}	5	5	5	1	15	15	80	1	15	5	15	1	10^4	12	5	10^7	-
Score (%)	0	6	0	0	6	13	61	0	27	100	0	0	58	42	86	88	**46**
			0.15				**0.34**				**0.47**			**0.5**		**0.87**	

affect the results of privacy indicators evaluation. For this case study, weights for each privacy indicator (i.e., inclination) are equally set (i.e., the value of 0.2 for each privacy indicator). However, if a user has a different preference on each privacy indicator, the weights can be changed that results different privacy score. For instance, when the values for the inclination indicator are set to have equal weights for all 16 sub-indicators, such that $W_{i\in\{1,2,\cdots,16\}} = \frac{1}{16}$, the evaluated scores with the given threshold set $\{\tau_{mild}, \tau_{standard}, \tau_{strict}\}$ respectively become {0.69, 0.60, 0.48} for Nest and {0.61, 0.49, 0.37} for Google Home.

The results of the evaluation of Nest and Google Home with respect to the anonymous users are provided in Table 3 and Table 4. According to the subjective indicator scores of the two users, the user of the Nest more frequently use the service for a longer period of time compared to the other user of the Google Home even though the general reputation of Google Home is quantitatively higher than that of Nest. This results demonstrate that how the score of a service can be evaluated differently considering the characteristics of each user, such as the usage patterns and sensitivity levels.

5. Conclusion

In this paper, a framework for privacy indicators evaluation of services that handle personal data is proposed. The privacy indicators are first defined in standardized terms, and the proposed framework is then investigated to evaluate the level of privacy risks and compliance of the service in terms of the defined privacy indicators. In this paper, we have focused on introducing a generalized privacy indicators evaluation framework rather than developing more advanced technologies for each method and work process in the evaluation. In particular, this paper proposes various criteria to evaluate and quantify objective and subjective privacy indicators for measuring a privacy score of services by analyzing privacy policies, list of requested data permissions, users' service usage statistics, etc. As a proof of concept of the proposed framework, more than 10,000 privacy policies of mobile applications and case studies on well-known mobile applications for smart home service are investigated based on the proposed framework. The case studies support the feasibility of systematical implementation and practical deployment of the proposed framework. In the meantime, the case studies in this paper only cover a few evaluation methods that can be applied to the proposed framework. For future works of this study, more advanced NLP technologies and machine learning models can be extensively applied with a wider range of evaluation criteria to improve evaluation performance. As the extension of this study, therefore, more advanced technologies for each part of the evaluation can be investigated, and the proof of concept provided in this paper can be further developed as a complete system for the service launch.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Ahmad, W., Chi, J., Tian, Y., Chang, K.-W., 2020. PolicyQA: A Reading Comprehension Dataset for Privacy Policies. In: Findings of the Association for Computational Linguistics: EMNLP 2020, Association for Computational Linguistics, pp. 743–749.

Bannihatti Kumar, V., Iyengar, R., Nisal, N., Feng, Y., Habib, H., Story, P., Cherivirala, S., Hagan, M., Cranor, L., Wilson, S., Schaub, F., Sadeh, N., 2020. Finding a Choice in a Haystack: Automatic Extraction of Opt-Out Statements from Privacy Policy Text. In: Proc. the Web Conf., WWW '20, pp. 1943–1954.

Barth, S., Ionita, D., de Jong, M.D.T., Hartel, P.H., Junger, M., 2021. Privacy rating: A user-centered approach for visualizing data handling practices of online services. IEEE Trans. Prof. Commun. 64 (4), 354–373.

Becher, S.I., Benoliel, U., 2021. Law in books and law in action: the readability of privacy policies and the GDPR. Consumer Law Econ.

California Consumer Privacy Act (CCPA), 2022. https://oag.ca.gov/privacy/ccpa/. Accessed: Jan. 14, 2022.

Chowdhury, G., 2010. Introduction to Modern Information Retrieval. Facet Publications, Facet.

Dagger, T.S., Danaher, P.J., Gibbs, B.J., 2009. How often versus how long: the interplay of contact frequency and relationship duration in customer-reported service relationship strength. J. Service Res. 11 (4), 371–388.

Disconnect Me, 2021. https://web.archive.org/web/20170709022651/disconnect.me/icons/. Accessed: Nov. 16, 2021.

Elallid, B.B., Benamar, N., Hafid, A.S., Rachidi, T., Mrani, N., 2022. A comprehensive survey on the application of deep and reinforcement learning approaches in autonomous driving. J. King Saud Univ.- Comput. Informat. Sci.

General Data Protection Regulation (GDPR), 2022. http://data.europa.eu/eli/reg/2016/679/oj. Accessed: Jan. 14, 2022.

Google Play Store. Google Home, 2022. https://play.google.com/store/apps/details?id=com.google.android.apps.chromecast.app. Accessed: Apr. 19, 2022.

Google Play Store. Nest. 2022. https://play.google.com/store/apps/details?id=com.nest.android. Accessed: Apr. 19, 2022.

Haring, C., Klaarwater, C., Bouwmans, G., Cools, B., Gurp, P., Meer, J., Postma, C., 2020. Validity, reliability and feasibility of a new observation rating tool and a post encounter rating tool for the assessment of clinical reasoning skills of medical students during their internal medicine clerkship. A pilot study. BMC Med. Educ. 20 (198), 1–7.

Harkous, H., Fawaz, K., Lebret, R., Schaub, F., Shin, K.G., Aberer, K., 2018. Polisis: Automated Analysis and Presentation of Privacy Policies Using Deep Learning. In: Proc. USENIX Conf. Security Symp., pp. 531–548.

ISO Central Secretary, 2017. Information technology – Security techniques – Privacy framework. Standard ISO/IEC 29100:2011, International Organization for Standardization, Geneva, Switzerland.

ISO Central Secretary, 2019. Security techniques – Extension to ISO/IEC 27001 and ISO/IEC 27002 for privacy information management – Requirements and guidelines. Standard ISO/IEC 27701:2019, International Organization for Standardization, Geneva, Switzerland.

ITU-T, 2017. Overview of trust provisioning for information and communication technology infrastructures and services. Standard ITU-T Y.3052, Telecommunication Standardization Sector of International Telecommunication Union (ITU-T), Geneva, CH.

ITU-T, 2020. Framework for trust-based personal data management. Standard ITU-T Y.3055, Telecommunication Standardization Sector of International Telecommunication Union (ITU-T), Geneva, CH, Sept.

Keymanesh, M., Elsner, M., Sarthasarathy, S., 2020. Toward domain-guided controllable summarization of privacy policies. In: Proc. Natural Legal Language Processing (NLLP) co-located with ACM Int. Conf. Knowledge Discovery & Data Mining (KDD), vol. 2645, pp. 18–24.

Linden, T., Harkous, H., Fawaz, K., 2020. The Privacy Policy Landscape After the GDPR. volume 1. sicendo.

Malgieri, G., Custers, B., 2018. Pricing privacy – the right to know the value of your personal data. Comput. Law Sec. Rev. 34 (2), 289–303.

Mccaffrey, D., Oliveri, M., Holtzman, S., 2018. A Generalizability Theory Study to Examine Sources of Score Variance in Third-Party Evaluations Used in Decision-Making for Graduate School Admissions: A G Theory Study for Third-Party Evaluations. ETS Res. Rep. Series 2018 (9).

Nejad, N.M., Graux, D., Collarana, D., June 2019. Towards Measuring Risk Factors in Privacy Policies. In: Proc. International Conf. AI and Law (ICAIL).

Nokhbeh Zaeem, R., Anya, S., Issa, A., Nimergood, J., Rogers, I., Shah, V., Srivastava, A., Barber, K.S., 2020. PrivacyCheck v2: A Tool That Recaps Privacy Policies for You. In: Proc. ACM Int. Conf. Information & Knowledge Management, ACM, pp. 3441–3444.

Oliner, A.J., Iyer, A.P., Stoica, I., Lagerspetz, E., Tarkoma, S., 2013. Carat: Collaborative Energy Diagnosis for Mobile Devices. In: Proc. ACM Conf. Embedded Netw. Sens. Syst., New York, NY, USA.

Oussous, A., Benjelloun, F.-Z., Ait Lahcen, A., Belfkih, S., 2018. Big data technologies: A survey. J. King Saud Univ.- Comput. Informat. Sci. 30 (4), 431–448.

Pandey, B., Kumar Pandey, D., Pratap Mishra, B., Rhmann, W., 2021. A comprehensive survey of deep learning in the field of medical imaging and medical natural language processing: Challenges and research directions. J. King Saud Univ.- Comput. Informat. Sci.

Poplavska, E., 2020. From prescription to description: Mapping the GDPR to a privacy policy corpus annotation scheme. Front. Artif. Intell. Appl. 334.

Privacy Icons, 2021. https://wiki.mozilla.org/privacy_icons/. Accessed: Nov. 16, 2021.

Ravichander, A., Black, A.W., Wilson, S., Norton, T., Sadeh, N., 2019. Question Answering for Privacy Policies: Combining Computational and Legal Perspectives. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Association for Computational Linguistics, pp. 4947–4958.

Ravichander, A., Black, A.W., Norton, T., Wilson, S., Sadeh, N., 2021. Breaking Down Walls of Text: How Can NLP Benefit Consumer Privacy? In: Proc. Annual Meeting of the Association for Computational Linguistics and Inter. Joint Conf. Natural Language Processing, pp. 4125–4140.

Reidenberg, J.R., Breaux, T., Cranor, L.F., French, B., Grannis, A., Graves, J.T., Liu, F., McDonald, A., Norton, T.B., Ramanath, R., Russell, N.C., Sadeh, N., Schaub, F., 2015. Disagreeable Privacy Policies: Mismatches Between Meaning and Users' Understanding. Berkeley Technol. Law J. 30 (1), 39–88.

Reidenberg, J.R., Russell, N.C., Herta, V., Sierra-Pambley, W., Norton, T., 2019. Trustworthy privacy indicators: grades, labels, certifications and dashboards. Washington Univ. Law Rev. 96 (6), 49–67.

Sathyendra, K.M., Schaub, F., Wilson, S., Sadeh, N. 2016. Automatic Extraction of Opt-Out Choices from Privacy Policies. In: Proc. Association for the Adv. Artificial Intell. (AAAI).

Schaub, F., Balebako, R., Durity, A.L., Cranor, L.F., 2015. A design space for effective privacy notices. In: Symposium On Usable Privacy and Security (SOUPS 2015), USENIX Association, pp. 1–17.

Story, P., Zimmeck, S., Ravichander, A., Smullen, D., Wang, Z., Reidenberg, J., Russell, N., Sadeh, N., 2019. Natural Language Processing for Mobile App Privacy Compliance. In: Proc. Association for the Adv. Artificial intell. (AAAI).

ToS;DR, 2021. https://tosdr.org/. Accessed: Nov. 16, 2021.

TrustArc, 2021. https://trustarc.com/. Accessed: Nov. 16, 2021.

Wilson, S., Schaub, F., Dara, A.A., Liu, F., Cherivirala, S., Giovanni Leon, P., Schaarup Andersen, M., Zimmeck, S., Sathyendra, K.M., Russell, N.C., Norton, T.B., Hovy, E., Reidenberg, J., Sadeh, N., 2016. The Creation and Analysis of a Website Privacy Policy Corpus. In: Proc. Annual Meeting of the Association for Comput. Linguistics (ACL), pp. 1330–1340.

Zimmeck, S., Wang, Z., Zou, L., Iyengar, R., Liu, B., Schaub, F., Wilson, S., Sadeh, N., Bellovin, S., Reidenberg, J., 2017. Automated Analysis of Privacy Requirements for Mobile Apps. In: Netw. and Distrib. Syst. Security (NDSS) Symp.

Zimmeck, S., Story, P., Smullen, D., Ravichander, A., Wang, Z., Reidenberg, J.R., Russell, N.C., Sadeh, N.M., 2019. MAPS: Scaling Privacy Compliance Analysis to a Million Apps. Proc. Privacy Enhancing Technol. 66–86, 2019.